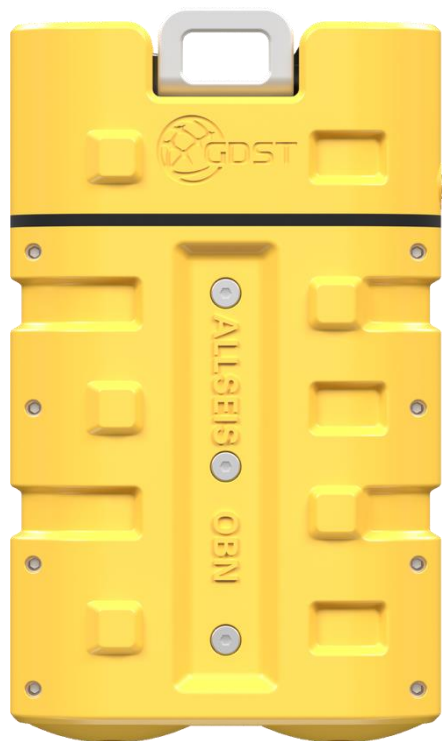




General Features

Geophone	14Hz omnidirectional high sensitivity geophone Optional 3Hz omnidirectional active force feedback wideband geophone
Hydrophone	3.4Hz wideband high sensitivity hydrophone
Digitalization	24-bit Δ - Σ ADC with GNSS disciplined clock
Storage capacity	64GB
Battery Life	Internal rechargeable Li-ion battery (300WH), over 1080 hours (45days) continuous recording
Timing Accuracy	+/- 1 μ s (continuous GNSS disciplined) +/- 2ms (OCXO within 30 days)
Automated Tests	Internal testing signal generator for function and performance self-inspection
LED Status Indicator	OBN health, GNSS time synchronization status
Data Harvest	Data download/battery charging unit
Weight	11kg (in air), 6.6kg (in water)
Size	35cm x 21cm x 11cm
Operating Temperature	-10°C ~ +45°C
Operating Depth	0~500m

Product Highlights

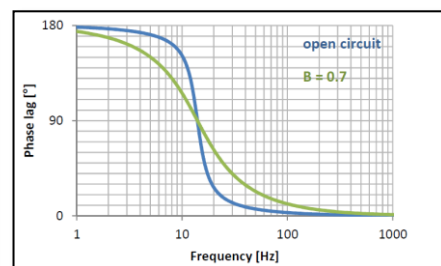
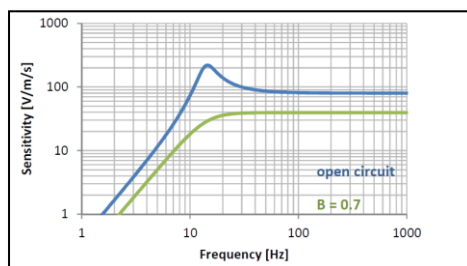
- Game changing 2-component OBN for high quality shallow water seismic acquisition down to 500m;
- All-in-one design supports either node-on-a-rope (NOAR) or remotely-operated vehicle (ROV);
- Built-in 14Hz omnidirectional high sensitivity geophone;
- Optional 3Hz active force feedback geophone for broadband seismic survey;
- Built-in standard 3.4Hz hydrophone;
- 24-bit Δ - Σ ADC converter for high resolution, high dynamic range seismic signal recording;
- Configurable sample rate and preamplifier gain;
- Built-in test signal generator supports full signal chain function and performance self-inspection;
- Built-in high precision thermostatic crystal (OCXO) for $\pm$ 2ms timing accuracy ;
- Compact volume (35L x 21W x 11H cm) and light weight (11kg) for convenient deployment and retrieval.

Acquisition Specifications

ADC Resolution	24-bit
Sampling Interval	0.5ms, 1ms, 2ms, 4ms
Preamplifier Gain	x1, x4, x16 (0dB, 12dB, 24dB)
Gain Accuracy	0.1%
Full Scale Input Signal	$\pm$ 2.5Vp-p @ x1 gain $\pm$ 625mVp-p @ x4 gain $\pm$ 156mVp-p @ x16 gain
Instantaneous DR (1ms, typical value)	125dB @ x1 gain
Full Scale DR	145dB
Equivalent Input Noise (typical value)	1 μ V @ x1 gain 0.25 μ V @ x4 gain
Harmonic Distortion	-120dB @ 31.5Hz
Frequency Response	0Hz ~ 866Hz @ 0.5ms (-3dB) 0Hz ~ 800Hz @ 0.5ms (-0.1dB)
Anti-alias Filter	Linear filter, 86.6% of Nyquist -3dB bandwidth
Stopband Attenuation	>120dB @Nyquist
Common Mode Rejection	>100dB
Inter Channel Rejection	>100dB

14Hz Omni-directional Geophone

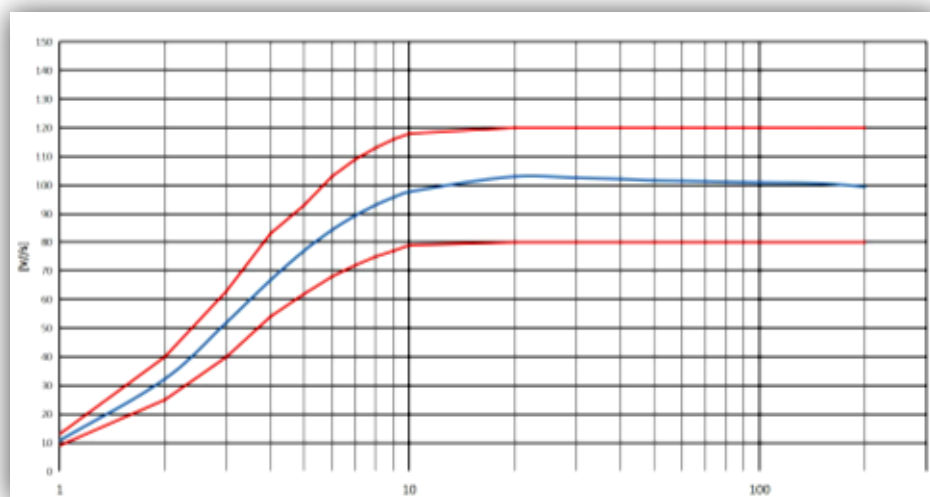
Natural Frequency	14Hz
Sensitivity	80V/m/s
Spurious	> 190Hz
Harmonic Distortion	≤0.1% horizontal, ≤0.4% vertical
Coil Resistance	3500Ω
Tilt	0°~360°



14Hz High Sensitivity Omni-directional Geophone Response Curve and Phase Lag

3Hz Omni-directional Wideband Geophone

Natural Frequency	3Hz
Sensitivity	100V/m/s
spurious	> 190Hz
Distortion	< 0.1% horizontal, < 0.4% vertical
Tilt	0°~360°
Equivalent Input Noise	1uVrms@3~190Hz
Full Scale DR	125dB



3Hz Omnidirectional Wideband Geophone Response Curve

Hydrophone

Frequency Response	3.4Hz@~3dB
Sensitivity	8.9V/bar
Working Depth	≤1000m
Operating Temperature	-10°C~+50°C

Stackable Charging /Downloading Unit

Slots	3
Charging Time	6 hours
Charging Power	350W
Unit Charging Power	100W
Full Download Rate	2GB/min.
Unit Download Rate	0.66GB/min.
Interface	1G Ethernet
Operating Temperature	0°C~+40°C
Size	1U

* Specifications or information subject to change without notice

